



Paper birch competitive effects vary with conifer tree species and stand age in interior British Columbia forests: implications for reforestation policy and practice

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Abstract

Paper birch (*Betula papyrifera* Marsh) is considered the most important competitive threat to commercially valuable conifers in the highly productive interior cedar–hemlock forests of British Columbia. It is routinely removed from conifer plantations at high cost to increase conifer growth rates and meet reforestation policy regulations. Competitive effects of paper birch and other neighbours on conifer growth were measured in mixed forests to test the hypotheses that the importance and intensity of interspecific competition were (a) high in young plantations; (b) decrease with stand age; (c) decrease with conifer species shade tolerance. Neighbourhood analysis was applied to 748 target conifer trees in 11 stands that included three stand age classes (11, 25, and 50 years-old) and three target conifer species (Douglas-fir (*Pseudotsuga menziesii* Mirb. Franco), western redcedar (*Thuja plicata* D. Don), and western larch (*Larix occidentalis* Nutt.)). All sites had similar site characteristics and disturbance histories. Regression models were used to examine the importance (model adjusted r^2) and intensity (model β_1) of neighbour competition on target conifer size. Double logarithmic regression models explained on average $38 \pm 3\%$ of the variation in target conifer diameter from the neighbourhood index, NRI (neighbour relative height index) = $\sum((\text{neighbour basal area} \times \text{height})/\text{target conifer height})$. The models indicated that broadleaf neighbours were more important and intense competitors in the 11- and 25-year-old stands, and conifers were more important and intense competitors in the 50-year-old stands. When all broadleaf and conifer neighbours were included in the models, the intensity and/or importance of neighbour competition decreased in the order of $25 > 50 > 11$ -year-old stands, and western larch $>$ Douglas-fir $>$ western redcedar. Instead of encouraging uniform broadleaf removal across most conifer plantations in the BC interior, reforestation policies should take into account the variable effects of broadleaf competition within stands, across stand ages, and with different tree species compositions.

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1. Introduction

Broadleaf trees are considered a serious competitive threat to regenerating conifers because of their rapid growth rate, high longevity and high resource demands (Wagner et al., 2001). In northern temperate forests, broadleaves are routinely removed from conifer plantations at considerable expense to reduce the negative effects of resource competition on conifer growth (Lieffers et al., 1996). Global opposition to herbicide use has led to increased use of repeated manual cutting and increased cost of removal treatments (Dohrenbusch and Frochot, 1993; Wagner, 1993), which commonly account for the largest portion of reforestation expenditures (BC Ministry of Forests, 2001). At the same time, the benefits of broadleaves on forest productivity and biodiversity are receiving increased attention (Burton et al., 1992), with several studies showing improved total yield in mixed versus pure conifer stands (Kelty, 1992; Sachs, 1995; Mård, 1996; Man and Lieffers, 1999; Frivold and Kolström, 1999; Valkonen and Valsta, 2001), protection of understorey conifers against damaging frost (Perala and Alm, 1990), improved habitat for cavity nesting birds (Aitken et al., 2002), and improved soil microbial diversity and function (Bradley and Fyles, 1995; Jones et al., 1997; DeLong et al., 2002). The ecological and economic benefits of managing landscapes that include mixtures of conifers and broadleaves as natural components of succession are being increasingly recognised in European and American forest plans and policies (Swanson et al., 1997; Valkonen and Valsta, 2001; Juha et al., 2002).

In interior British Columbia, several conifer species commonly associate with broadleaved trees, forming stratified, single cohort, mixed species stands (Cameron, 1996). Western redcedar (*Thuja plicata* D. Don), interior Douglas-fir (*Pseudotsuga menziesii* var. *glauca* (Beissn.) Franco), and western larch (*Larix occidentalis* Nutt.), which are described as shade tolerant, moderately shade tolerant, and shade intolerant, respectively (Klinka et al., 2000; Carter and Klinka, 1992), regenerate alongside intolerant broadleaves in the interior wet belt forests following stand-replacing fires (Johnson et al., 1990) or clearcutting (Simard and Vyse, 1994). The density and proportion of regenerating species vary widely depending on seed source, microsite conditions, and degree of distur-

bance, with localised broadleaf densities ranging between 1000 and 60,000 stems ha⁻¹ up to 10 years post-disturbance (Simard and Vyse, 1994). Species variation in height growth and shade tolerance results in early dominance by broadleaves and intolerant conifers, but intense vertical stratification quickly gives rise to mixed conifer–broadleaf stands, with eventual dominance by shade tolerant conifers such as western redcedar and western hemlock (*Tsuga heterophylla* (Raf.) Sarg.) (Cameron, 1996).

Early in stand development, most paper birch (*Betula papyrifera* Marsh), trembling aspen (*Populus tremuloides* Michx.) and black cottonwood (*Populus balsamifera* L. ssp. *trichocarpa* T. & G.) trees are routinely removed from plantations to ensure the successful regeneration of conifers (Comeau et al., 2000). Broadleaf removal has been uniformly applied in many areas regardless of stand age or species composition (Simard et al., 2001), primarily because complete removal greatly reduces the costly risk of not meeting reforestation policy standards. Reforestation policies and regulations remain conifer-biased and are too restrictive to allow variable levels of broadleaf retention under different conditions. This partly arises from the historically lucrative conifer market in North America, but also from the large literature extolling the benefits of intensive silviculture (Lavender et al., 1990).

Recently, a variety of extensive silviculture practices have been proposed for northern temperate forests that would allow development of mixed stands with a range of composition and structural values, resulting in diverse, semi-natural landscapes that also include intensively managed conifer plantations (Lieffers et al., 1996; Lautenschlager, 2000; Greene et al., 2002). These practices have broad applicability to managing mixed forests of aspen, birch and conifers in boreal and temperate forests across North America and Europe (Fries et al., 1997; McCarthy, 2001; Sendak et al., 2003; Kabzems and Garcia, 2004). To develop effective silviculture prescriptions for mixed species stands, however, foresters must understand competitive relations among species. This is not a simple task since vegetation competitive effects on conifers have been shown to vary with stand developmental stage (Burton, 1993; Newton and Jolliffe, 1998a; Wagner, 2000), species composition, and site quality (Simard, 1990a; Burton, 1993; Coates and

Burton, 1999). The identity of important competitors has been shown to shift with succession, for example, from herbs and shrubs soon after disturbance to neighbouring trees at later stages (Harrington et al., 1995; Wagner and Radosevich, 1998; Newton and Jolliffe, 1998b). Shade tolerance has also influenced conifer growth response to neighbour abundance (Carter and Klinka, 1992; Coates and Burton, 1999), where each species reacts differently according to shade tolerance patterns and autecology (Drever and Lertzman, 2001). Shade tolerant species like western redcedar, for example, increase growth rates rapidly but reach an asymptote quickly with increasing light availability, whereas shade intolerant species increase growth rates linearly (Wright et al., 1998). In keeping with this, conifer growth responses have been shown to increase with increasing species shade intolerance following broadleaf removal and increased light availability (Simard et al., 2001).

Given the variability of conifer response to competition, the range of desired stand compositions, and the cost of manually controlling broadleaf trees, it is important that foresters manage broadleaf competition with an understanding of when and where it significantly reduces conifer growth. Understanding the effects of stand age and species composition on the nature of broadleaf competition should lead to improved vegetation management prescriptions that are better suited to local site conditions, and to improved predictions of treatment effects on stand productivity and structure (Comeau et al., 1999; Coates and Burton, 1999; Drever and Lertzman, 2001). Evaluating changes in competition as forest stands age, however, is hindered by the shortage of long-term manipulative competition experiments (Comeau et al., 2000). Many studies that have examined competitive effects have been short-term, limiting our understanding and ability to predict outcomes of treatments applied at different stages of stand development (Burton, 1993).

This study sought to characterise the growth response of western redcedar, interior Douglas-fir, and western larch to a range of broadleaf and conifer competition levels. We characterised differences in competitive interactions among three stand ages: 11, 25 and 50 years-old. We also examined differences in competitive effects among conifer species that varied widely in shade tolerance and growth rate. Specific

hypotheses were that: (1) competition, particularly broadleaf competition, significantly reduces growth of all target conifer species; (2) the importance and intensity of competition decrease with increasing stand age; (3) the importance and intensity of competition decrease with increasing conifer species shade tolerance.

2. Methods

2.1. Study sites

This study was undertaken in the mixed conifer–broadleaf ‘wet belt forests’ of southern interior British Columbia. Sampling occurred in two variants of the interior cedar–hemlock biogeoclimatic zone (ICHmw2, ICHmw3) and one variant of the interior Douglas-fir zone (IDFmw2) (Lloyd et al., 1990). The area has warm, moist summers and moderately cold, snowy winters. Mean temperature during the growing season is 14–16 °C, mean minimum temperature in January is –7 to –12 °C, and mean annual precipitation is 520–670 mm, of which 220–290 mm falls as rain during the growing season.

Eleven mixed forest sites were selected randomly from a list of suitable sites generated by silviculture treatment history and forest inventory records in the BC Ministry of Forests. The age classes of the stands were 8–14 (11 ± 3) years, 23–27 (25 ± 2) years, and 45–55 (50 ± 5) years. In the 11-year age class, four replicate sites were sampled for each conifer species (Douglas-fir, western redcedar, and western larch). In the 25- and 50-year age classes, there were three replicate sites with Douglas-fir and western redcedar, and one with western larch.

All of the 11-year-old and two of the 25-year-old sites were plantations that had been logged and prepared with prescribed fire, and the third 25-year-old site and all of the 50-year-old sites were native forests that had been burned by wildfire. Individual sites were 1–2 ha in size, and varied in slope (5–35%) and aspect. Soils on all sites were deep, well-drained, of loamy sand to sandy loam texture, and derived from glacial moraine parent material. The soils were classified either as Humo-Ferric Podzols or Dystric Brunisols (Soil Classification Working Group, 1998). All sites were located between 50°30'N and 51°45'N, and

Table 1
Stand, site history and neighbourhood characteristics of the sample sites

Stand	Age ^a by species ^b (years)	Site history ^c	Neighbourhood density (stems ha ⁻¹)		Neighbourhood mean height (m)	
			Broadleaf	Conifer	Broadleaf	Conifer
11-Year-old stands						
First	13 Fd and Cw; 13–15 Ep	L72, B83, P83Fd	1699 (180)	3711 (298)	6.2 (0.1)	2.5 (0.0)
Rachel's	10 Lw and Ep	L84–85, B86, P86Lw	5192 (719)	875 (55)	4.0 (0.1)	3.7 (0.2)
Release	13 Fd, Cw, Lw; 11–13 Ep	L78, B80, P83Fd	10451 (1088)	3269 (201)	4.2 (0.1)	3.3 (0.0)
Heaven	9 Fd, Cw, Lw; 8–9 Ep	L86, B86, P87Fd	4509 (257)	1617 (110)	4.1 (0.2)	2.2 (0.0)
Larch Lane	12 Fd, Cw, Lw; 14 Ep	L79, B80, P84Fd	4357 (309)	5048 (368)	5.2 (0.1)	2.7 (0.0)
25-Year-old stands						
67 sign	23 Fd, Cw, Ep	WF69–70, P73Fd	3495 (230)	4677 (285)	7.0 (0.1)	3.7 (0.0)
Backup	25 Fd, Cw, Lw, Ep	L67, B67–70	2105 (139)	3028 (140)	5.6 (0.1)	4.2 (0.1)
Excavator	23 Fd, Cw, Lw, Ep	L72, B72	2681 (173)	3966 (307)	7.0 (0.1)	3.0 (0.0)
50-Year-old stands						
Eyeball	50 Fd, Cw, Ep	WF46	526 (28)	2982 (96)	13.4 (0.1)	4.6 (0.0)
Reneker Ck	50–55 Fd, Cw, Ep	WF39	209 (17)	1948 (60)	14.4 (0.1)	6.6 (0.2)
Wishing	45–55 Fd, Cw, Lw, Ep	WF39	183 (8)	1261 (37)	12.3 (0.2)	7.8 (0.1)

^a Age at the time of sampling as determined by history records and coring.

^b Conifer species paired with paper birch. Species codes: Fd = Douglas-fir; Cw = western redcedar; Lw = western larch.

^c Site history codes include an activity code followed by the year of the activity and, in the case of planting, the planted species. The activity codes are: L = logged; B = broadcast burned; P = planted; WF = wildfire.

118°46'W and 119°32'W. Elevations ranged from 650 to 1000 m. Site index (50 years) of paper birch on all sites ranged between 25 and 30 m, and all sites was classified as medium in moisture and nutrient regime.

All stands were dominated by paper birch, Douglas-fir, western redcedar and, where possible, western larch. Other conifers typical of wet belt stands were also usually present at low densities. These included western hemlock, lodgepole pine (*Pinus contorta* Dougl.), western white pine (*Pinus monticola* Dougl.), interior spruce (*Picea engelmannii* Parry *x glauca* (Moench) Voss), and sub-alpine fir (*Abies lasiocarpa* (Hook.) Nutt.). The 11-year-old stands ranged in density between 5400 and 13,700 stems ha⁻¹ (3.1–8.7 m² ha⁻¹), and averaged 8146 stems ha⁻¹ (5.9 m² ha⁻¹) (Table 1, Fig. 1). Dominant trees reached 11–15 m height and 9–13 cm diameter, depending on the site. Paper birch was the most common and largest species (67% of stand basal area), dominating the upper canopy in all five stands. The 25-year-old stands ranged in density between 5100 and 8200 stems ha⁻¹ (8.2–13.0 m² ha⁻¹), and averaged 6651 stems ha⁻¹ (10.5 m² ha⁻¹). Dominant trees, which tended

to be western larch and Douglas-fir, reached 17–22 m height and 17–20 cm diameter, depending on the site. Paper birch was still the most abundant species on average (43% of stand basal area), followed by Douglas-fir (20%) and western redcedar (17%). There was some evidence of paper birch mortality in standing and downed woody debris. The 50-year-old stands ranged in density between 1400 and 3500 stems ha⁻¹ (9.9–11.6 m² ha⁻¹), and averaged 2370 stems ha⁻¹ (11.0 m² ha⁻¹). Dominant trees, which were consistently Douglas-fir and larch, reached 28–30 m height and 30 cm diameter. Paper birch and other broadleaves comprised a relatively small portion of the stand (23% of stand basal area). Paper birch occurred only in the intermediate and co-dominant height classes, having been surpassed by longer-lived conifers in the upper canopy and excluded from the lower canopy by shade. There was considerable evidence of birch mortality in standing and downed wood.

Composition of the shrub and herb communities varied among stand ages. The most common shrub and herb species in the 11-year-old stands included thimbleberry (*Rubus parviflorus* Nutt.), falsebox

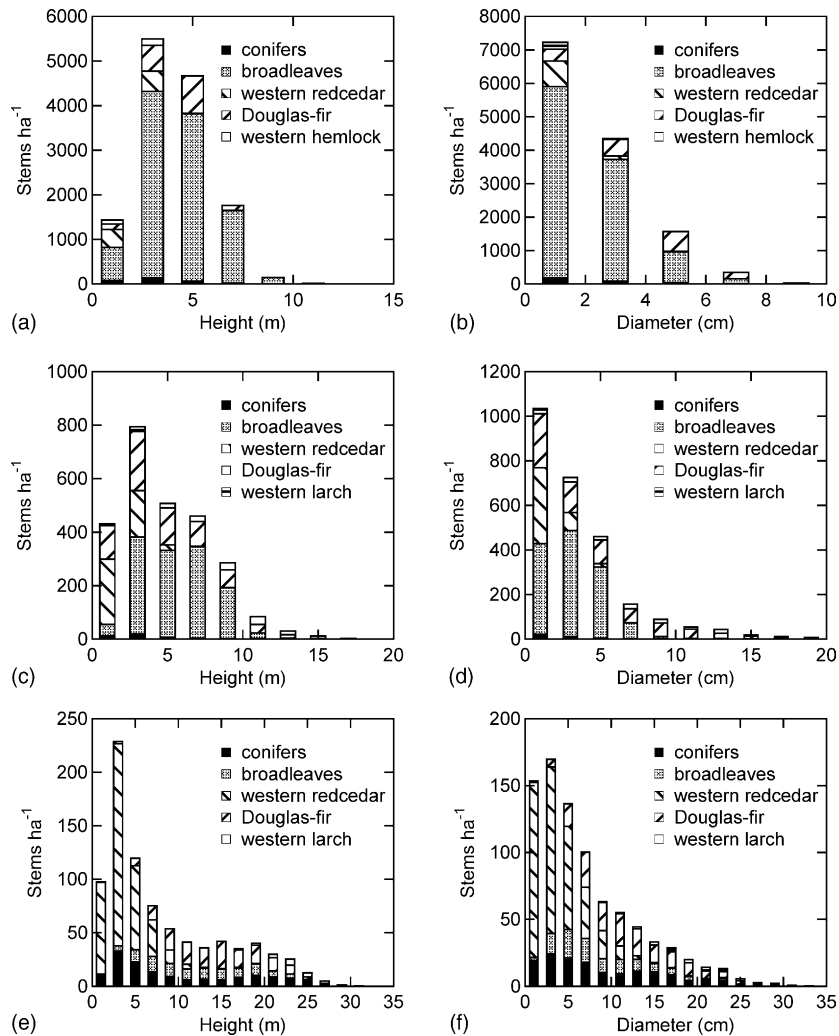


Fig. 1. Height and diameter distributions by species or species group for one example site in each of the 11-, 25- and 50-year age classes, where the sites are Release (a, b), Backup (c, d), and Wishing (e, f).

(*Pachistima myrsinites* (Pursh) Raf.) and fireweed (*Epilobium angustifolium* L.). In the 25- and 50-year-old stands, the dominant shrub and herb species were thimbleberry, falsebox, buffalo berry (*Shepherdia canadensis* (L.) Nutt.), prickly rose (*Rosa acicularis* Lindl.), birch-leaved spirea (*Spirea betulifolia* Pallas), fireweed, and wild sarsparilla (*Aralia nudicalus* L.).

2.2. Sampling design

A density gradient of paper birch neighbours was established for each target tree species by:

(1) defining three sampling strata based on the range in paper birch density (stems ha⁻¹) that naturally occurred on the site (strata were identified for low (<33% of maximum), medium (33–66% of maximum) and high (>66% of maximum) birch density); (2) systematically locating sample plots in a grid until up to 10 target trees per conifer species were selected for each sampling stratum (approximately 30 plots per species). This sampling design captured the widest range of birch densities around individual target conifer trees that occurred on each site. The study included 748

target conifer trees distributed among the 11 study sites.

The closest undamaged target tree of each species to each grid-point was selected as the sample tree, and a neighbourhood plot was centred upon each target tree stem. Neighbourhood plots had a radius of 3.99 m (0.005 ha) in the 11-year-old sites, 7.1 m (0.016 ha) in the 25-year-old sites, and 15.0 m (0.071 ha) in the 50-year-old sites. Plot sizes were chosen to ensure that at least 15 neighbours occurred in maximum density plots and that they were large enough to include full paper birch clumps. These criteria, plus the fact that stem density decreased with stand age, resulted in larger plot sizes in the older age classes. Overlap between adjacent neighbourhood plots for a particular target conifer species was minimised through selection of appropriate plot intervals, but this was not a concern for different target trees species located near the same grid-point because their competitive relationships with neighbours were examined separately.

2.3. Conifer measurements

Target conifers were measured in the 11-year-old sites in June–August 1995, and in the 25- and 50-year-old sites in June–August 1996 and 1997. Measurements were diameter (diameter outside bark at 1.3 m above ground), diameter increment, and total height. Two-year diameter increment in the 11-year-old stands was calculated by subtracting 1997 diameter from 1995 diameter. Five-year diameter increment in the 25- and 50-year-old stands was measured on cores taken at 1 m height. Diameter and height measurements were used to calculate height:diameter ratio (total height/diameter) and estimate stem volume using the formula for a cone ($(\pi \times (\text{diameter}/2)^2 \times \text{total height})/3$).

2.4. Neighbourhood measurements

Neighbourhood measurements were taken at the same time that target conifer trees were measured, when leaf area was maximised for the growing season. Neighbourhoods were quantified using two approaches. The first approach grouped all plants by species or species group in the circular plot centred on the target conifer stem. For each broadleaf and conifer tree species, and each dominant shrub and herb spe-

cies, percent crown cover (visually estimated to the nearest 5%) above and below the height of the target conifer tree was recorded. To qualify as a dominant shrub or herb, the species must have occurred in at least 80% of the site plots. Species were also grouped into four groups (broadleaf trees, conifer trees, shrubs, herbs), and percent cover greater than and less than the height of the target conifer tree recorded. The second approach considered only individual broadleaf and conifer tree neighbours. In each plot, diameter, total height, and species of every broadleaf and conifer tree were recorded. These measures were used to calculate basal area ($\pi \times (\text{diameter}/2)^2$) and density (number of individuals) per species or species group per plot.

2.5. Neighbourhood competition indices

Seven neighbourhood abundance indices (NAI's) were developed. Two indices were measures of neighbour abundance at the species and species-group level: (1) percent cover above the height of the target conifer; (2) percent cover below the height of the target conifer. Five indices were measures of neighbour abundance at the species group level. For the broadleaf, conifer, and broadleaf + conifer tree groups only, these included: (1) basal area; (2) height; (3) basal area $\times$ height; (4) number of trees; (5) number of trees $\times$ height, each summed for all trees in the neighbourhood. In addition to the neighbour abundance indices, a neighbour abundance relative to tree size index (NRI) was tested:

$$\text{NRI} = \frac{\text{NAI}}{\text{height}} \quad (1)$$

where height is the total height of the target conifer tree. Using this equation, the single NAI estimate per tree group for a plot was divided by the total height of the target conifer tree.

2.6. Neighbourhood model development

The analysis focused on developing simple regression models that integrated neighbourhood competition indices (independent variables) in a manner that accounted for the greatest variation in target conifer tree size and growth (dependent variables). Multiple regression was not used because (i) several independent variables shown important to tree performance in

other studies either were held constant or not measured in this study (Simard, 1989; Wagner and Rado-sevich, 1991a); (ii) other studies have shown that it usually does not lead to a consistent model applicable to all sites (Brand, 1986; Simard, 1990b); (iii) it was shown to be of limited value in an earlier study on similar sites by Simard (1990b).

The response variable best predicted by neighbour indices (i.e. highest adjusted r^2) the most frequently across all target conifer species and stand ages was diameter. Diameter increment, total height, height:diameter ratio, and stem volume index were commonly predicted to the same degree, but this did not

consistently occur in all sites. For these reasons, and for brevity, we present only diameter results.

The most appropriate functional forms between the response variables and each neighbourhood competition index were identified. Following sample site-specific graphical assessment of the trends between each response variable and neighbourhood competition index, three candidate functional forms were tested: linear, semi-logarithmic and double logarithmic. The linear association between the functional forms of each response variable and competition index was evaluated on the basis of the distribution of residuals, constancy of variances, adjusted r^2 , and

Table 2

Slope (β_1) and adjusted r^2 for regression models predicting target conifer diameter from NRI's that included broadleaves only, conifers only, and all broadleaf and conifer trees

Stand age class (years)	Target conifer species	Sample site	All		Broadleaf		Conifer		
			β_1	Adjusted r^2	β_1	Adjusted r^2	β_1	Adjusted r^2	
11	Western redcedar	First	-0.22	0.25	-0.38	0.37	-0.11	0.07	
		Heaven	-0.16	0.03	-0.25	0.13	-0.02	0.03	
		Larch Lane	-0.72	0.37	-0.56	0.42	0.01	0.00	
		Release	-0.29	0.23	-0.03	0.05	-0.32	0.14	
	Douglas-fir	First	-0.13	0.05	-0.14	0.14	-0.09	0.03	
		Heaven	-0.02	0.10	-0.06	0.13	0.00	0.00	
		Larch Lane	-0.42	0.28	-0.25	0.27	-0.25	0.12	
		Release	-0.48	0.47	-0.31	0.37	0.02	0.00	
	Western larch	Rachels	-0.51	0.27	-0.38	0.25	-0.04	0.00	
		Heaven	-0.13	0.16	-0.12	0.16	-0.01	0.05	
		Larch Lane	-0.63	0.45	-0.43	0.38	-0.44	0.16	
		Release	-1.07	0.55	-0.60	0.33	-0.53	0.29	
25	Western redcedar	67 sign	-0.81	0.57	-0.63	0.59	-0.44	0.18	
		Backup	-0.85	0.17	-0.14	0.03	-0.46	0.11	
		Excavator	-0.94	0.62	-0.56	0.50	-0.49	0.31	
	Douglas-fir	67 sign	-0.68	0.66	-0.36	0.30	-0.57	0.63	
		Backup	-0.62	0.23	-0.37	0.37	-0.09	0.01	
		Excavator	-0.88	0.72	-0.61	0.61	-0.31	0.17	
	Western larch	Backup	-0.53	0.57	-0.35	0.31	-0.31	0.36	
	50	Western redcedar	Eyeball	-0.65	0.57	-0.77	0.72	-0.29	0.24
			Reneker	-0.72	0.57	-0.14	0.06	-0.58	0.52
Wishing			-0.34	0.39	-0.10	0.15	-0.33	0.36	
Douglas-fir		Eyeball	-0.68	0.26	-0.36	0.32	0.02	0.00	
		Reneker	-0.39	0.27	-0.07	0.07	-0.32	0.21	
		Wishing	-0.72	0.58	-0.01	0.00	-0.58	0.51	
Western larch		Wishing	-0.65	0.49	-0.11	0.17	-0.61	0.43	

Models are shown for each conifer target species in each sampled stand. The general form of the model was $\log_e(\text{diameter}) = \log_e \beta_0 + \beta_1 \log_e(\text{NRI})$, where $\text{NRI} = \sum((\text{basal area} \times \text{height})/\text{target conifer height})$. All models were significant at $P < 0.01$.

the significance of the correlation ($P < 0.01$). Following this analysis, the double logarithmic model was selected:

$$\log_e(y) = \log_e(\beta_0) + \beta_1 \log_e(\text{NRI or NAI}) \quad (2)$$

where y = diameter, β_0 and β_1 are equation-specific intercept and slope parameters estimated by least-squares regression analysis for each target conifer species and site, and NRI or NAI are the neighbour competition indices. For logarithmic transformations of NRI or NAI, observations with a value of zero were assigned a value of 0.0001, which was chosen based on the smallest measurements. In regression models relating conifer size to percent cover of neighbours, the best fit was found using untransformed variables.

Among the competition indices, NRI measures were substantially more predictive of target conifer diameter than NAI, with adjusted r^2 values averaging 0.28 and 0.07, respectively, across all sites (Table 2, NAI data not shown). The focus of this paper is therefore on NRI. By comparing regression results among all NRI's tested as described above, we found the one most frequently resulting in the highest adjusted r^2 was $\Sigma(\text{basal area} \times \text{height}) / \text{conifer height}$. Based on our results, we focus this paper on this best NRI. Species groups (e.g. all broad-leaf trees species, all conifer tree species) were better predictors of target conifer diameter than individual species, and tree species groups were far better predictors than shrubs or herbs (for brevity, results from individual species, shrub and herb regressions are not shown).

The regression models were used to evaluate the importance and intensity of neighbour competition on target conifer performance. Weldon and Slauson (1986) have shown that the slope of the regression (β_1) can represent the intensity of competition and the coefficient of determination (adjusted r^2) can represent its importance. Patterns in adjusted r^2 values were examined to compare the importance of neighbour competition among age classes and target conifer species. The intensity of neighbour competition (model β_1) was compared among age classes and target conifer species using two-way analysis of variance (ANOVA) (after Newton and Jolliffe, 1998a). The extra sums of squares principle was used to test whether full regression models that included separate slope and intercept terms for each age class or conifer

species had a better fit than simple models with only one slope and one intercept (Draper and Smith, 1981). Indicator variables were used to remove age class or conifer species terms from the full model and an F -test was used to test whether there were significant changes to the sums of squares.

3. Results

3.1. Effect of competition on conifers

Neighbour competition, considered over whole stands, accounted for a relatively low amount of variation in conifer diameter for all target tree species and stand ages studied. This is partly because tree size was generally highly variable under low to moderate levels of competition and was constrained only by high levels of competition in localised, dense broad-leaf patches (Fig. 2).

Scattergraphs of the fitted regressions models relating conifer diameter with NRI are shown for each target conifer species and stand age class in Fig. 3. In each stand and species group, all five NRI measures were consistently either negatively or positively correlated with conifer diameter, indicating consistent effects of neighbours on conifer performance. Where the neighbour group included broadleaves only or broadleaves and conifers together, the NRI was consistently negatively correlated with conifer diameter

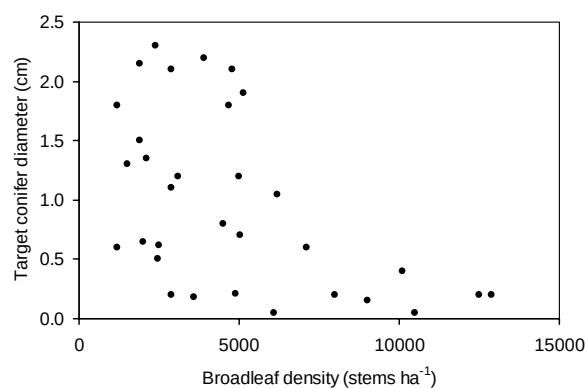


Fig. 2. Scattergraph showing relationship between target conifer diameter and density of all broadleaf neighbours (stems per hectare) on one example site. In this example, the target conifer species is western redcedar at the Larch Lane site in the 11-year age class.

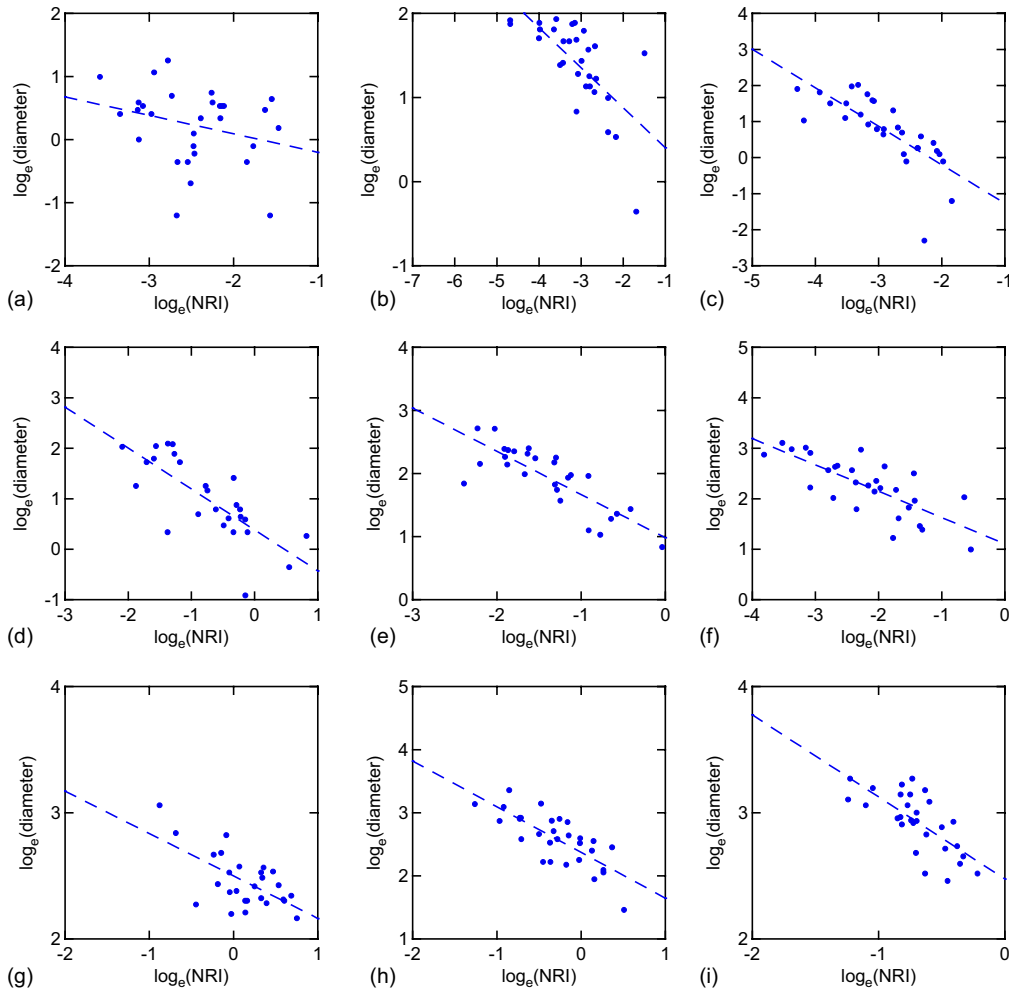


Fig. 3. Relationship between diameter of target conifers ($\log_e(\text{diameter})$) and the relative neighbour size index ($\log_e(\text{NRI})$) for one example stand for each target conifer species and stand age class. The sites are 11-year-old Release (a–c), 25-year-old Backup (d–f), and 50-year-old Wishing (g–i), where target conifer species are shown in the order of western redcedar (a, d, g), Douglas-fir (b, e, h) and western larch (c, f, i).

on all sites. Where only conifers were included in the index, the NRI was also usually negatively correlated with target conifer diameter, but there was no correlation on three 11-year-old sites and one 50-year-old site (Table 2). The NRI including both broadleaves and conifers was used in most subsequent analyses of stand age and target species effects because it explained more variation in target conifer diameter than the indices that included broadleaves alone or conifers alone. Adjusted r^2 values where only broad-leaf neighbours were included in the NRI averaged

0.28 compared with 0.38 where all neighbours were included (Table 2).

Percent cover of broadleaves or conifers that were taller or shorter than target conifers was generally weakly correlated with target conifer diameter (Table 3). Percent cover was a better predictor of conifer diameter than either the sum of basal area or the sum of basal area $\times$ height, however. Percent cover of taller broadleaves and taller conifers was generally negatively correlated with conifer diameter (Figs. 4a and 5a), whereas percent cover of shorter

Table 3

Slope (β_1) and adjusted r^2 for regression models predicting target conifer diameter from percent cover indices that included taller or shorter broadleaves or conifers

Stand age class (years)	Target conifer species	Sample site	Broadleaves PCT		Broadleaves PCS		Conifers PCT		Conifers PCS	
			β_1	Adjusted r^2	β_1	Adjusted r^2	β_1	Adjusted r^2	β_1	Adjusted r^2
11	Cedar	First	−0.01	0.03	0.10	0.01	−0.02	0.13	0.06	0.29
		Heaven	−0.02	0.07	0.22	0.14	0.00	0.00	0.04	0.04
		Larch Lane	−0.01	0.06	0.14	0.04	0.02	0.04	0.05	0.20
		Release	0.00	0.02	0.02	0.03	−0.02	0.13	0.09	0.16
	Douglas-fir	First	−0.02	0.03	0.20	0.24	−0.04	0.19	0.06	0.31
		Heaven	−0.01	0.01	0.13	0.04	−0.06	0.13	0.14	0.16
		Larch Lane	−0.01	0.02	0.19	0.32	−0.02	0.02	0.06	0.25
		Release	−0.05	0.36	0.06	0.10	−0.04	0.07	0.11	0.26
	Western larch	Heaven	−0.03	0.05	0.05	0.07	−0.08	0.04	0.13	0.21
		Larch Lane	−0.09	0.25	0.24	0.78	−0.15	0.24	0.14	0.16
		Release	−0.05	0.15	0.07	0.25	−0.12	0.38	0.20	0.64
25	Cedar	67 sign	0.00	0.00	0.44	0.72	−0.04	0.07	0.16	0.58
		Backup	0.00	0.00	0.14	0.28	−0.02	0.07	0.05	0.29
		Excavator	−0.07	0.14	0.30	0.13	−0.19	0.51	0.12	0.24
	Douglas-fir	67 sign	−0.08	0.15	0.20	0.25	−0.16	0.38	0.08	0.10
		Backup	−0.11	0.44	0.08	0.13	−0.06	0.09	0.17	0.52
		Excavator	−0.03	0.12	0.29	0.10	−0.06	0.19	0.03	0.03
	Western larch	Backup	−0.32	0.43	0.14	0.15	−0.25	0.51	0.27	0.36
50	Cedar	Eyeball	0.05	0.05	0.46	0.33	−0.04	0.04	0.12	0.62
		Reneker	0.03	0.03	0.74	0.13	−0.07	0.24	0.07	0.26
		Wishing	−0.06	0.08	0.31	0.34	−0.07	0.18	0.02	0.01
	Douglas-fir	Eyeball	−0.16	0.33	0.21	0.40	−0.18	0.19	−0.03	0.02
		Reneker	−0.12	0.29	0.20	0.31	−0.16	0.26	0.15	0.18
		Wishing	−0.14	0.19	0.11	0.12	−0.24	0.66	0.04	0.03
	Western larch	Wishing	−0.28	0.31	0.03	0.01	−0.19	0.48	−0.04	0.03

The general form of the model was: diameter = $\beta_0 + \beta_1$ (PCT or PCS), where PCT is percent cover equal to or taller than the target conifer and PCS is percent cover shorter than the target conifer. All models were significant at $P < 0.01$.

broadleaves and shorter conifers was positively correlated (Figs. 4c and 5c), regardless of target species or age class. Some of the trends among age class and species that occurred in the NRI models also occurred in the percent cover models: (1) the amount of variation in conifer diameter explained by percent cover (taller broadleaves or conifers) tended to be greater in the 25- and 50-year-old sites than the 11-year-old sites (Figs. 4b and 5b); (2) taller conifers explained slightly more variation in conifer diameter than taller broadleaves (Figs. 4b and 5b); (3) intensity of competition with broadleaves or conifers was greater in the 25- and 50- than the 11-year-old site Figs. 4a and 5a); (4)

intensity (Figs. 4a and 5a) and importance (Figs. 4b and 5b) of competition from taller broadleaves or conifers decreased in order of western larch >> Douglas-fir > western redcedar, regardless of stand age.

3.2. Effect of stand age on competition

The amount of variation in target conifer diameter explained by NRI tended to increase with stand age (Fig. 6). Even though this trend was identified with an NRI that included broadleaves and conifers together, we found that broadleaves alone explained more

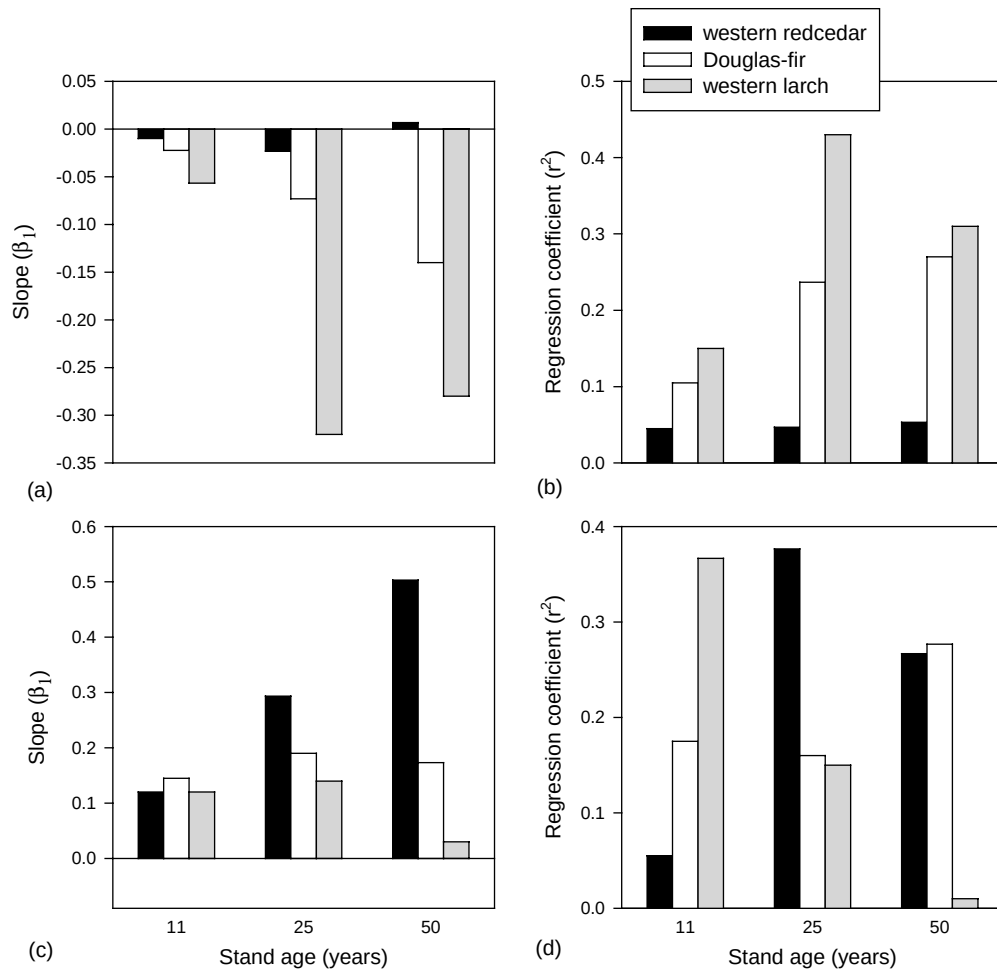


Fig. 4. Mean slope (β_1) and adjusted r^2 for regression models predicting target conifer diameter from percent cover indices of taller (a, b) and shorter (c, d) broadleaves. The general form of the model was: diameter = $\beta_0 + \beta_1$ (PCT or PCS), where PCT is percent cover equal to or taller than the target conifer and PCS is percent cover shorter than the target conifer. All models were significant at $P < 0.01$.

variation with a greater competitive intensity than conifers alone on the 11-year-old sites, whereas the reverse was generally true for the 50-year-old sites (Fig. 6c–f). This corresponded with a lower proportion of broadleaves and corresponding greater proportion of conifers in the older stands. We also found that competition intensity (β_1) between neighbours (all broadleaves and conifers) and target conifers was significantly greater on the 25- than the 11-year-old sites and was intermediate on the 50-year-old sites ($P = 0.0182$, Fig. 6a). The extra sums of squares analysis showed the same significant effect of age class. Removing either separate intercept or slope

terms for each age class from the full model significantly reduced the sums of squares ($P < 0.0001$, Table 4), indicating stand age significantly influenced competition intensity between neighbours and target conifers.

3.3. Effect of target conifer species on competition

The importance and intensity of neighbour competition varied among target conifer species. On 11-year-old sites, the importance of neighbour competition (adjusted r^2) tended to be similar for western redcedar and Douglas-fir but was greater for western

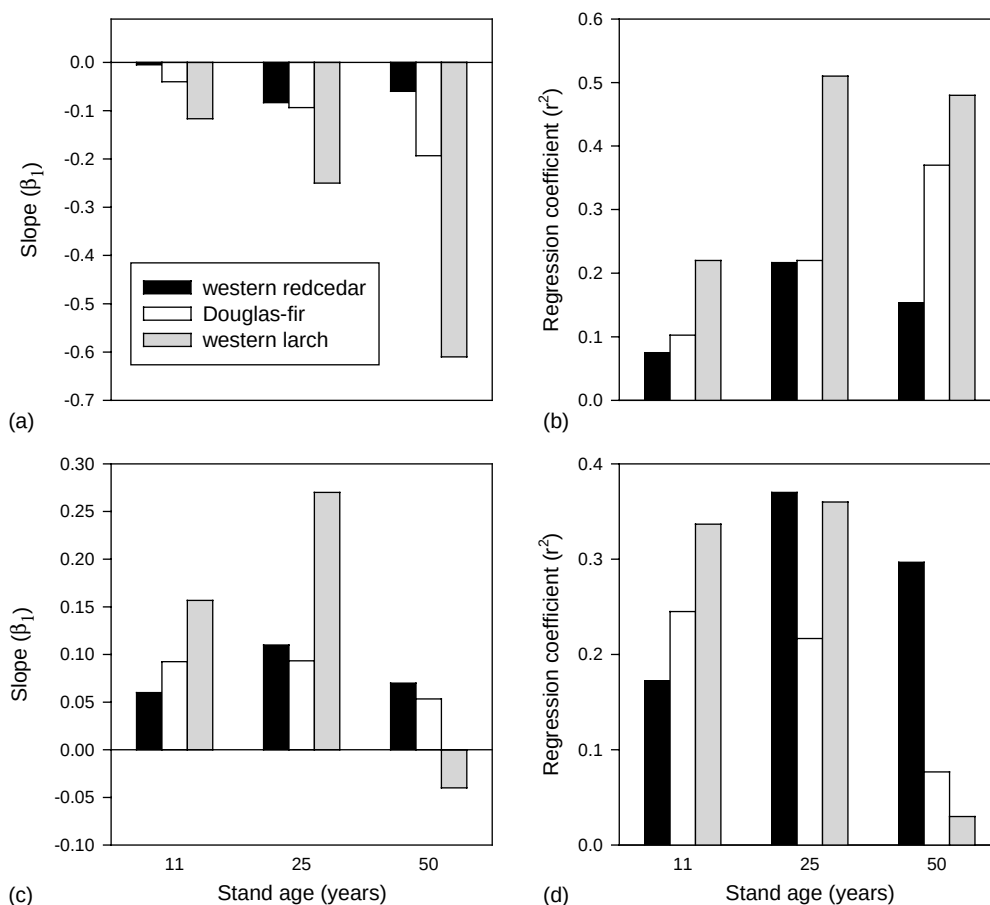


Fig. 5. Mean slope (β_1) and adjusted r^2 for regression models predicting target conifer diameter from percent cover indices of taller (a, b) and shorter (c, d) conifers. The general form of the model was: diameter = $\beta_0 + \beta_1$ (PCT or PCS), where PCT is percent cover equal to or taller than the target conifer and PCS is percent cover shorter than the target conifer. All models were significant at $P < 0.01$.

larch (Fig. 6b). Competition intensity (β_1) followed the same trend, but differences among species across all age classes were not significant ($P = 0.4441$, Fig. 6a). Intensity was also similar for western redcedar and Douglas-fir on 25- and 50-year-old sites (there were too few replicates to evaluate western larch) (Fig. 6a). The same result was found using the extra sums of squares principle, where removing each species significantly affected the full model with separate intercepts ($P < 0.0001$) but not slopes ($P = 0.6891$, Table 4). Target conifer species differed in size at zero competition but did not differ significantly in their response to increasing levels of competition.

4. Discussion

4.1. Does broadleaf competition reduce target conifer growth?

We found a generally weak relationship between conifer size and broadleaf competition in the commonly found, low neighbour densities. This agrees with other findings that variation in juvenile tree growth increases below thresholds of competition or above certain levels of resource availability (Wang et al., 1994; Burton, 1996; Wagner, 2000; Drever and Lertzman, 2001). Although the regression model adjusted r^2 values were generally low, the slopes were

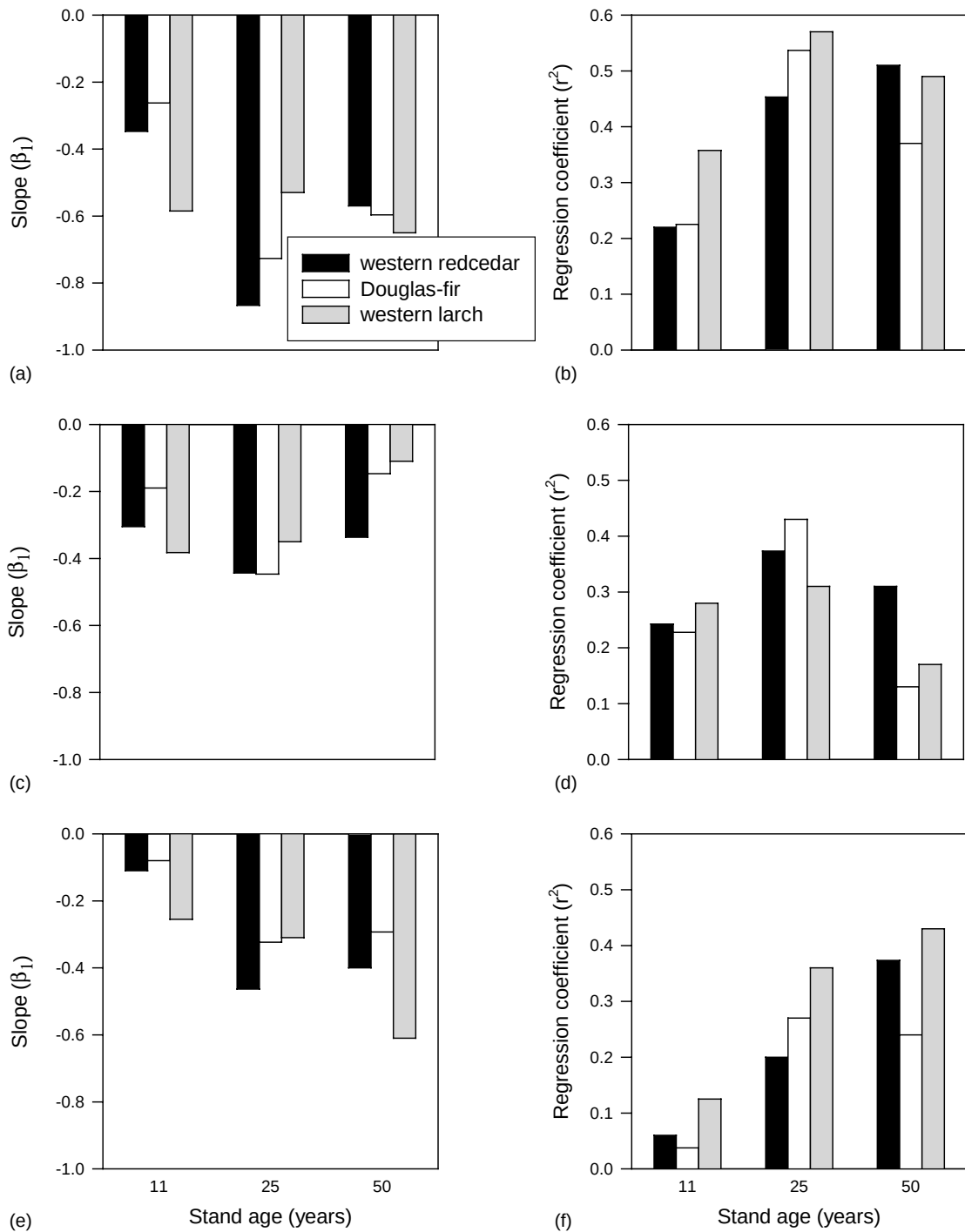


Fig. 6. Mean slope (β_1) and adjusted r^2 for regression models predicting target conifer diameter from NRI's that included all broadleaf and conifer trees (a, b), broadleaves only (c, d) and conifers only (e, f). The general form of the model was $\log_e(\text{diameter}) = \log_e \beta_0 + \beta_1 \log_e(\text{NRI})$, where $\text{NRI} = \sum((\text{basal area} \times \text{height})/\text{target conifer height})$. All models were significant at $P < 0.01$.

Table 4

Extra sums of squares test results for effects of age class and target conifer species terms on the full model relating target conifer diameter to NRI (including both conifer and broadleaf neighbours)

Factor	Model	Variables in the model ^a	SSE	F-test	P-value
Age	Separate intercepts, separate slopes	Cw β_0 , Fd β_0 , Lw β_0 , Cw β_1 , Fd β_1 , Lw β_1	327.71		
	Common intercept, separate slopes	β_0 , Cw β_1 , Fd β_1 , Lw β_1	719.03	439.43	<0.0001
	Common slope, separate intercepts	β_1 , Cw β_0 , Fd β_0 , Lw β_0	708.83	427.98	<0.0001
	Common intercept, common slope	β_0 , β_1	886.20	313.58	<0.0001
Species	Separate intercepts, separate slopes	Cw β_0 , Fd β_0 , Lw β_0 , Cw β_1 , Fd β_1 , Lw β_1	633.49		
	Common intercept, separate slopes	β_0 , Cw β_1 , Fd β_1 , Lw β_1	768.99	78.71	<0.0001
	Common slope, separate intercepts	β_1 , Cw β_0 , Fd β_0 , Lw β_0	634.13	0.37	0.6891
	Common intercept, common slope	β_0 , β_1	886.20	73.40	<0.0001

The significance (F-test) of separate intercept (β_0) and slope (β_1) parameters for different age and species terms was determined by the reduction in SSE.

^a Variables are denoted by species code and regression coefficient. Species codes are provided in Table 1.

sometimes steep, suggesting that competition was detectable and locally intense, but only moderately important overall (Weldon and Slauson, 1986; Burton, 1993).

The best index of interspecific competition in our study was based on tree neighbour basal area and height, and a relative measure of target conifer height, explaining on average $38 \pm 3\%$ of the variation in target conifer diameter. This contrasts with the neighbour abundance indices, which explained only $7 \pm 1\%$ of the variation in target conifer diameter. Indices that include a relative size term have been used by others to account for influences of tree size and past performance on tree response to current competitive conditions, and have explained up to 98% of the variation in target conifer growth in shrubby and herbaceous plant communities (Lorimer, 1983; Brand, 1986; Daniels et al., 1986; Comeau et al., 1993). The lower r^2 values in our mixed broadleaf stands may have resulted from a patchier, less intensively competitive environment for well-established sapling conifers than the more uniformly intensive competitive environment commonly experienced by seedlings recently planted into lush herb-dominated communities (Simard et al., 2001). Instead of using diameter as the dependent variable, others have suggested using recent growth because it would better reflect the current competitive environment without the confounding effects of historical environments (Brand, 1986; Newton and Jolliffe, 1998a). However, we found that NRI accounted for a similar amount of variation in diameter as diameter increment in each stand.

Visual estimates of percent cover were more predictive tree neighbour abundance measures than the more objective measures of basal area or stems per hectare, explaining $20 \pm 2\%$ of the variation in target conifer diameter. Percent cover of taller trees was generally negatively correlated with target conifer diameter whereas percent cover of shorter trees was positively correlated, possibly because shorter trees were replacing larger, more competitive trees. Percent cover of herbs and shrubs was poorly correlated with target conifer diameter, which was expected given their low abundance and small size relative to the target conifers. Small neighbours usually have a smaller competitive effect because they take up fewer resources (Harper, 1977). Visual cover estimates of neighbours have been shown to be highly correlated with leaf area (Wagner and Radosevich, 1991b), and have provided the most precise estimates of interspecific competitive effects with conifer species in shrubby and herbaceous plant communities (Coates, 1987; Simard, 1990a; Wagner and Radosevich, 1998). Wagner and Radosevich (1998) suggest that visual cover estimates may integrate some important attribute of neighbour canopies (e.g. foliage density and vigour) that is missed by the more objective measures of plant abundance. In older tree-dominated communities, however, neighbour measures of basal area, volume or biomass have had more common usage (Weiner, 1984; Newton and Jolliffe, 1998a).

Our results agree with previous studies showing it is difficult to accurately predict individual tree growth rate at a given neighbourhood competition level (Brand,

1986; Coates, 1987; Simard, 1989; Burton, 1993). This is because variation in growth of individual trees can result from many factors other than competition (Drever and Lertzman, 2001; Kozłowski, 2002), including genetics (St. Clair and Snieko, 1999), predation, damage or disturbance (Perry, 1985), disease (Simard et al., 2001), initial seedling size (Wagner and Radosevich, 1991a), previous periods of suppression (Wright et al., 1998), interactions with soil organisms (Simard et al., 1997), or microsite quality (Goldberg et al., 1999; Arian and Turkington, 2001). Our target conifers were likely influenced by some of these factors, particularly in the older stands, which were naturally regenerated rather than planted (Coates and Burton, 1999).

4.2. Does competition vary with stand age?

We found that the importance and intensity of tree neighbour competition on target conifers differed considerably among stand age classes, reflecting the dynamic nature of seral mixed forests. In models predicting conifer diameter from NRI, the importance and intensity of competition peaked in the 25-year-old stands. This is because the 25-year-old stands had reached full site occupancy (high basal area) yet still had high stocking density (high stems per hectare). Stands with these characteristics are in the stem exclusion stage of stand development (Oliver and Larson, 1996), and they experience intense inter-tree competition, begin self-thinning, and have maximal leaf area (Kozłowski, 2002). The 50-year-old stands were also in stem exclusion, but greater mortality had occurred and inter-tree competition was of lower intensity than in the 25-year-old stands. By contrast, the 11-year-old stands were not yet fully occupied; competition was not as fierce or as important in determining individual tree growth, both features that characterise the initiation stage of stand development (Oliver and Larson, 1996).

Identity of the most important and intense competitors also differed among stand age classes. Broadleaves were the most important competitors in the 11- and most of the 25-year-old stands, whereas conifers were more important in most of the 50-year-old stands. This shift in interspecific competition corresponded with differences in broadleaf–conifer composition and stand structure among age classes (Hara and Wyszomirski, 1994), mainly as a result of self-thinning (Kozłowski, 2002). Composition changed

from 75% broadleaf/25% conifer in the 11-year-old stands, to 50/50% and 23/77% in the 25- and 50-year-old stands, respectively. Stands also stratified vertically based on species differences in growth patterns and shade tolerance, resulting in a shift in dominance of the upper canopy from primarily broadleaf at 11 years to mixed broadleaf–conifer at 25 years and primarily conifer at 50 years. Observations of coarse woody debris suggested that most competition-related mortality that had occurred in the 25-year-old stands was among broadleaves, whereas both broadleaves and some conifers had died in the 50-year-old stands. The broadleaves behaved as pioneer species while the conifers survived in a subordinate crown position when the stands were young; when birch started to die out in the older stands, the conifers differentiated further into upper crown classes based on their shade tolerance (Oliver and Larson, 1996), agreeing with earlier observations of stand development in this forest type (Cameron, 1996). Our results are in line with Scandinavian studies that show shifts in competitive effects of birch (*B. pubescens* Ehrh. and *B. pendula* Roth) with stand age (Frivold and Kolström, 1999). Frivold and Frank (2002) found that stand growth was better in young mixed stands of Norway spruce (*Picea abies* (L.) Karst.) and birch than in pure spruce stands because of differences in species growth patterns and more effective utilisation of resources (Mård, 1996). This trend disappeared in older stands because of natural declines in birch growth rates (Frivold and Frank, 2002). Shifts in competition have also been measured in shrub and herb communities over much shorter time spans because of natural succession (Harrington et al., 1995; Wagner and Radosevich, 1998) and in black spruce stands of similar age sequences as our stands because of spruce budworm defoliation (Newton and Jolliffe, 1998b).

4.3. Does competition vary among conifer species?

Neighbour competitive effects varied among conifer species, reflecting differences in their tolerance to resource limitations and position in the canopy. In the 11-year-old stands, the importance and intensity of neighbour competition tended to be greatest for western larch and were similar for Douglas-fir and western redcedar. These results agree with mixed stand studies in Scandinavian forests, where shade intolerant Scots

pine (*Pinus sylvestris* L.) was more negatively affected by increasing birch competition than tolerant Norway spruce (*Picea abies* (L.) Karst.) (Frivold and Kolström, 1999; Frivold and Frank, 2002). The interspecific trends we detected were comparable with results of other field studies in British Columbia investigating light-growth responses and shade tolerance (Klinka et al., 2000; Wang et al., 1994; Wright et al., 1998), which rank our species in order of increasing shade tolerance as western larch < Douglas-fir < western redcedar. Growth of shade tolerant species like western redcedar are less affected by shading from neighbours than shade intolerant species like western larch, which grow much faster at low competition and high light levels (Kobe and Coates, 1997). The similarity we found between western redcedar and Douglas-fir contrasts with the different light response patterns measured for those species in coastal British Columbia (Drever and Lertzman, 2001), and may have resulted from differences in the range of light availability studied, regional climatic influences on the study sites, and the lower shade tolerance of coastal versus interior varieties of Douglas-fir (Wright et al., 1998). In the older stands, the intensity of neighbour competition was also similar for western redcedar and Douglas-fir, but neighbours became slightly more important determinants of 50-year-old western redcedar diameter. Competition from neighbours may have been more important to western redcedar growth during this period because it was increasingly suppressed in the understorey, whereas Douglas-fir was growing into an upper, less crowded canopy position.

The changes we observed in importance of neighbours to different tree species with stand age appear to be associated with shifts in target tree canopy position and light environment (Givnish, 1988), and subtle species shifts in shade tolerance. In young stands, western redcedar appears to require shade to establish whereas western larch and Douglas-fir are intolerant of shade (Krajina et al., 1982). The probability of early mortality under low light, for example, is much lower for western redcedar than for shade intolerant species growing at the same rate (Kobe and Coates, 1997). Burton (1993) showed similar patterns of shade tolerance at establishment, where 2-year-old western redcedar grew faster than Douglas-fir under the same densities of fireweed. As trees age, however, the efficiency at which light energy is absorbed changes

(Field, 1983) and hence tolerance of shade can change. Western redcedar, for example, responds well to increasing light availability when it reaches sapling size (Drever and Lertzman, 2001), and interior Douglas-fir becomes moderately shade tolerant (Klinka et al., 2000). Although western redcedar continues to be more shade tolerant than Douglas-fir as stands age (Carter and Klinka, 1992), its similar response to neighbour competition in our study may have been affected by its deeper and more shaded position in the broadleaf canopy.

4.4. Study limitations

The stands sampled in this study represented an age sequence of mixed stand development (Sprugel, 1984). There are shortcomings to the use of chronosequences, however, including the possibility of dissimilar historical or environmental conditions among different-aged sites (Crowell and Freedman, 1994). We controlled this as much as possible through careful selection of uniform sites with similar characteristics. There were stand history differences, however, in that the younger stands generally had regenerated following clearcut logging and prescribed fire, whereas the older ones had regenerated following wildfire. It is possible that this resulted in different initial densities of broadleaf trees and conifers among sample sites. In addition, the neighbour density gradient within each site may have had an underlying environmental cause that also affected target conifer performance directly. This may have contributed to the low adjusted r^2 values of our regression models. For these reasons, competition relationships derived in this study should be tested in further studies where competition levels are experimentally manipulated. This said, there is no other way for forest managers to gain information for guiding decisions in young stands today. We recommend that managers use our results, but use them with caution, monitoring results, leaving controls, and generally following the adaptive management approach (Taylor et al., 1997).

4.5. Implications for reforestation policy and practice

Competition levels in young conifer stands are regulated by reforestation policy and associated

free-growing standards in British Columbia and other Canadian provinces (BC Ministry of Forests, 2000; Lieffers et al., 2002). In British Columbia, stands are assessed for free-growing (i.e. competition status) 5–15 years after establishment, and must be satisfactorily stocked with healthy, free-growing conifers of acceptable species. Conifers are considered free-growing when they are tall enough that they are not impeded by competition from neighbouring vegetation, and they will remain unimpeded in the future. The free-growing standard assumes that vegetation competition is primarily for light, that competition is the primary limitation on stand productivity, and that achievement of the standard ensures that stand productivity reflects site resources (Brand and Weetman, 1986).

Prior to 2000, a free-growing conifer could have no broadleaf tree greater than two-thirds its height within a 1 m radius (BC Ministry of Forests, 1995). Since then, the standards have been relaxed, and now allow some neighbouring broadleaves within a 3.99 m radius provided they are shorter than the conifer in three of four quadrants (BC Ministry of Forests, 2000). In the interior cedar–hemlock forests, two broadleaves are allowed for western larch, three for interior Douglas-fir, and five for western redcedar. In spite of this relaxation in the standards, almost none of the conifer plantations in the interior cedar–hemlock forests meet free-growing requirements without a brushing treatment to remove the broadleaf trees. To our knowledge, these brushing treatments commonly involve almost complete broadleaf removal so that the costly risk of not meeting free-growing is minimised. The biodiversity and productivity benefits associated with mixtures (Man and Lieffers, 1999; Valkonen and Valsta, 2001; Frivold and Kolström, 1999), however, suggest that free-growing standards ought to be designed to ensure maintenance of mixed species stands in the managed landscape. This may help avoid the biodiversity problems that are currently facing intensively managed forests in northern Europe (Fries et al., 1997; McCarthy, 2001; Lofman and Louki, 2001; Juha et al., 2002).

Our results suggest that broadleaf competition is only one of several factors influencing conifer productivity in the cedar–hemlock forests of southern interior British Columbia, accounting for only a moderate amount of the variation in conifer growth pat-

terns. While broadleaf removal should improve average conifer growth, depending on conifer species, treatment method and site quality (Simard et al., 2001), our results suggest that growth responses will be highly variable given the important influence of other non-competitive factors. Conifers growing well in spite of failing to meet free-growing may not respond to broadcast brushing treatments because some other factor imposes a greater limitation on their growth.

Forest managers often prescribe treatments when stands are young in an effort to prevent competition in the future. In interior British Columbia, for example, broadleaved trees are removed at an early age, even though they pose no immediate threat to conifer growth. Treatments are justified on the basis that, in time, the broadleaves will come to dominate the stand and thus reduce conifer productivity in the future. The perceived risk of not meeting free-growing is that conifer plantations will fail and turn into broadleaf stands. Our results suggest that this is highly unlikely, given that intensity and importance of competition with broadleaves decrease with stand age while competition with other conifers increases. This perception is also not supported by a landscape pattern analysis of recent vegetation change across 4.2×10^6 hectares in interior British Columbia, where deciduous cover remained relatively unchanged between 1975 and 1992 in spite of 8.5% disturbance to the forested area (Sachs et al., 1998). Although reductions in broadleaf density at an early age may reduce the intensity of competition that conifers experience as stands develop, our data suggests that most conifers will naturally outgrow broadleaves regardless of intervention.

Forest managers are accustomed to using shade tolerance to help guide regeneration decisions. Our results suggest that western larch is more affected by broadleaf competition than Douglas-fir or western redcedar because of its greater shade intolerance, supporting the current free-growing standard. While shade intolerant species should be given a higher priority in brushing decisions, assessments of shade tolerance should be based upon local observations. Shade tolerance of interior Douglas-fir appears to vary with interior regional climate, for example, and is greater than that of coastal Douglas-fir (Klinka et al., 2000).

4.6. Recommendations

In the interior cedar–hemlock zone, where most plantations contain mixtures of species that vary widely in shade tolerance, we recommend that managers employ selective treatments that focus primarily on release of tree species that are shade intolerant. We also recommend that treatment guides employ simple measures of competition, such as cover, rather than more complex methods. Finally we recommend that free-growing policy shift from a focus on relative vegetation stature at the individual tree level to stand level performance. This will reduce unnecessary brushing of well-growing trees, and will allow increased retention of broadleaf trees in mixed stands. Reducing the extensive treatment of most plantations would free up financial resources that could instead be focused on selective treatments where they are most needed, and on management of a diverse landscape that includes a healthy mosaic of conifer, broadleaf and mixed stands.

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